

“THE CZOCHRALSKI METHOD Is Preferred For Wafers Over 203mm”

How are large-diameter silicon wafers produced, and why does the Czochralski technique remain the industry's choice? Rajasekar Elavarasan of Raana Semiconductors explains to EFY's Nitisha Dubey how this method ensures the crystal precision essential for advanced IC fabrication.

Which technique is essential for manufacturing silicon wafers?

The Czochralski technique is the standard process for producing silicon ingots used in wafer manufacturing. The process begins with polysilicon as the raw material. Polysilicon is melted in a crucible at about 1500°C. A seed crystal is then dipped into the molten silicon and slowly pulled upward, allowing the material to solidify into a single-crystal ingot. These ingots, typically 300mm in diameter and up to 3-metres long, are then cut, lapped,

and polished to form silicon wafers. These wafers serve as substrates for microelectronic-grade IC fabrication, a process crucial to the production of modern electronic components.

Why is the Czochralski method preferred for producing silicon wafers?

The Czochralski method dominates large wafer production because it can produce silicon ingots exceeding 203mm (8-inches) in diameter, unlike the Bridgman or float-zone methods, which are limited to smaller sizes.

What are the key techniques for producing advanced semiconductors?

The Czochralski method is preferred for wafers over 203mm, while Float Zone or Inductive Load Zone techniques suit smaller wafers. Compound semiconductors, such as silicon carbide, are in increasing demand, particularly for electric vehicles, and are produced through the Physical Vapour Transport method. Gallium oxide, though limited to 50mm wafers, also holds potential. In quantum computing, sapphire substrates enable faster photonics-based IC communication—up to ten times quicker than electronic signals. These advancements reflect the shift toward high-performance semiconductors and quantum technologies.

How does each technique ensure crystal quality and the overall quality of semiconductor materials?

Each technique comes with its advantages and limitations. The Float Zone method, for instance, eliminates the use of a crucible, minimising contamination, since the material never contacts any external surface. This technique produces highly pure and efficient silicon ingots and wafers, but it is limited to a maximum diameter of 203mm. While this constraint makes it unsuitable for larger applications, it is ideal for sectors such as defence and aerospace, which prioritise material efficiency over size. In contrast, the Czochralski method



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